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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

QUESTION BANK

Regulation	:	2023
Course Name	:	AGILE SOFTWARE DEVELOPMENT
Course Code	:	JAL2006
Year/Semester/Branch	:	II/IV/AI&ML
Unit No	:	I
Unit Title	:	FUNDAMENTALS OF AGILE PROCESS
No. of. Hours	:	9
Weightage in Exam(%)	:	20

PART A

Q. No	Question	Marks	BTL	CO
1.	Mention the four core values of the Agile Manifesto? The four core values of the Agile Manifesto are: 1. Individuals and interactions over processes and tools. 2. Working software over comprehensive documentation. 3. Customer collaboration over contract negotiation. 4. Responding to change over following a plan.	2	K2	CO1
2.	State the primary focus of Agile methodologies? The primary focus of Agile methodologies is delivering value to the customer through iterative development, collaboration, and flexibility to change.	2	K1	CO1
3.	List the three main roles in Scrum? The three main roles in Scrum are: 1. Product Owner – Responsible for managing the product backlog and ensuring that the team works on the highest-priority items. 2. Scrum Master – Ensures that the Scrum process is followed, removes impediments, and facilitates communication. 3. Development Team – A self-organizing team responsible for delivering the increment of the product at the end of each sprint.	2	K2	CO1



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4.	Indicate a frequency of Sprint in Scrum? A Sprint is a time-boxed iteration, usually lasting between 1 and 4 weeks, in which a Scrum team works to complete a predefined set of product backlog items.	2	K1	CO1
5.	Give the purpose of the Scrum Board? The Scrum Board is a visual tool used to track the progress of tasks in a sprint, typically divided into columns like "To Do," "In Progress," and "Done."	2	K1	CO1
6.	Summarize the Pair Programming in XP? Pair programming is a practice where two developers work together on the same code, with one writing code (the driver) and the other reviewing it and providing guidance (the observer).	2	K2	CO1
7.	Define Test-Driven Development (TDD) in XP? Test-Driven Development (TDD) is a practice where developers write automated tests before writing the corresponding code, ensuring that the code meets the required functionality and passes the tests.	2	K1	CO1
8.	State the purpose of Refactoring in XP? Refactoring is the practice of continuously improving the design of the code by making small, incremental changes to improve readability, maintainability, and performance without altering its functionality.	2	K1	CO1
9.	Mention the main goal of Feature Driven Development (FDD)? The main goal of FDD is to deliver tangible, working software in the form of client-valued features, using a well-defined, step-by-step process for consistent and predictable progress.	2	K2	CO1
10.	Define the term "Feature" in FDD? In FDD, a feature is a small, client-valued functionality, typically taking one to two weeks to design and build, that can be delivered in a fully operational state.	2	K1	CO1
11.	How does FDD ensure that features are delivered on time? FDD ensures timely delivery by breaking down development into small, manageable features, each with a clear, two-week timeline for design and implementation, and regular progress tracking.	2	K2	CO1
12.	Why is frequent releases a key practice in XP? Frequent releases allow the team to deliver small, working pieces of the product quickly, gather feedback from the customer, and adjust the course of development accordingly.	2	K2	CO1
13.	Describe the main goal of Kanban?	2	K2	CO1



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	The main goal of Kanban is to optimize the flow of work, improve efficiency, and minimize bottlenecks by visualizing the process, limiting WIP, and focusing on incremental improvements.			
14.	What is a Kanban Board? A Kanban Board is a visual tool that displays the workflow of tasks, typically organized in columns such as "To Do," "In Progress," and "Done," to track work and improve visibility.	2	K2	CO1
15.	Discuss the purpose of limiting Work in Progress (WIP) in Kanban? Limiting WIP ensures that teams do not take on too many tasks at once, which helps improve focus, reduces multitasking, and enhances the flow of work, preventing bottlenecks.	2	K2	CO1
16.	Compare Kanban from Scrum? Kanban is more flexible, focusing on continuous delivery and flow with no fixed iterations, while Scrum is an iterative framework with fixed-length sprints and defined roles. Kanban emphasizes flow and limits WIP, whereas Scrum focuses on delivering work in chunks (sprints).	2	K2	CO1
17.	Mention the role of a "Kanban Board" in Lean Software Development? A Kanban Board is a visual tool used in Lean to track work items as they move through the development process. It helps teams visualize the flow of work, identify bottlenecks, and limit WIP to improve efficiency.	2	K1	CO1
18.	Define "Value Stream Mapping" in Lean Software Development? Value Stream Mapping is a technique used to analyze and visualize the flow of information and materials in the development process, helping identify inefficiencies, bottlenecks, and areas for improvement.	2	K1	CO1
19.	Classify the seven principles of Lean Software Development? The seven principles of Lean Software Development are: 1. Eliminate Waste – Remove anything that doesn't add value to the customer. 2. Build Quality In – Ensure quality is integrated at every stage of the process. 3. Create Knowledge – Continuously improve by generating knowledge and learning.	2	K2	CO1



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	4. Defer Commitment – Make decisions at the last responsible moment to maintain flexibility. 5. Deliver Fast – Deliver small increments quickly to get customer feedback and make improvements. 6. Respect People – Empower teams to make decisions and improve processes. 7. Optimize the Whole – Focus on improving the entire value stream, not just individual parts.			
20.	Interpret how Lean principles from manufacturing apply to software development Lean Software Development is an agile methodology that focuses on delivering value by eliminating waste, improving flow, and empowering teams to make decisions. It is inspired by Lean manufacturing principles, particularly those used in the Toyota Production System.	2	K2	CO1



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PART B

Q. No	Question	Marks	BTL	CO
1.	Demonstrate the Agile principles in detail and explain how they help in managing dynamic project requirements	16	K3	CO1
2.	A software company is developing a dynamic application where customer requirements frequently change. To handle these changes efficiently, the team adopts the Scrum framework. Explain in detail about scrum framework model. Analyze the advantages and limitations of the Scrum model.	16	K2	CO1
3.	Discuss the principles of Lean Software Development and explain how each principle helps in improving software development efficiency.	16	K2	CO1
4.	Determine how a Kanban board is used in software development and discuss how each principle contributes to improving the software development process.	16	K3	CO1
5.	Organization is working on a large-scale enterprise application with complex requirements and multiple teams. To manage the project efficiently, the company adopts Feature Driven Development (FDD), focusing on delivering small, client-valued features in a structured manner while maintaining quality and progress tracking. Apply the key concepts of Feature Driven Development (FDD) and explain its processes in detail.	16	K3	CO1
6.	Explain in detail about the key features of Extreme Programming.	16	K2	CO1

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PART A

Q. No.	Question	Marks	BTL	CO
1.	Why is a Product Vision important in Agile development? A Product Vision is important in Agile development because it aligns the team with the overarching goal and purpose of the product. It helps prioritize features and makes sure everyone works toward a shared understanding of the product's value	2	K2	CO2
2.	Mention the responsible employee for creating the Product Vision? The Product Owner is usually responsible for creating and maintaining the Product Vision, although collaboration with stakeholders, customers, and the development team is critical to ensure its alignment and relevance.	2	K1	CO2
3.	State the role of the Product Vision in stakeholder communication? The Product Vision serves as a communication tool to keep stakeholders aligned and engaged. It ensures that everyone, including customers, investors, and team members, understands the product's goals and the value it aims to deliver.	2	K1	CO2
4.	Discuss when Retrospectives be conducted in an Agile process? Retrospectives should be conducted at the end of every sprint or iteration, ideally during the last part of the sprint review, so that the team can reflect on their work and implement improvements in the next sprint.	2	K2	CO2
5.	State who can participates in a Retrospective? All members of the Agile team, including the Scrum Master, Product Owner, and development team members, typically participate in a Retrospective.	2	K1	CO2
6.	Define is a Retrospective in Agile? A Retrospective is a meeting held at the end of an iteration or sprint where the team reflects on what went well, what	2	K2	CO2



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	could be improved, and what actions can be taken to improve future performance.			
7.	Define Scrum Artifacts? Scrum Artifacts are key documents or tools that provide transparency and ensure everyone in the Scrum Team has a shared understanding of the work. These include the Product Backlog, Sprint Backlog, and Increment.	2	K1	CO2
8.	How to improve team collaboration? • Ensure that every team member is aligned with the vision and task at hand. • Include a varied skillset that does not overlap too much with other team members' specialties. • Establish and use regular feedback loops that are built into the learning cycle. • Avoid members of an Agile team being members of another team.	2	K2	CO2
9.	How Cross-functionality in Agile team will work? Cross-functional Agile teams are often compared to T-shaped ones, and they are similar. But team members who are cross-functional are more likely to possess skills outside of traditional disciplines. It's this wide-ranging skillset that lets them contribute to various projects, which in turn increases an Agile team velocity	2	K2	CO2
10.	What is an Agile team? An Agile team is a group of individuals with different multipurpose skills who are dedicated to the success of an Agile project through all stages of development, testing, and delivery. Agile teams are usually made up of 5–10 people who have been carefully selected for their expertise in specific business areas.	2	K1	CO2
11.	Illustrate the role of Coaches in XP Team. • Coaches help the team start their process well by arranging for a shared workspace and making sure that the team includes the right people. • They help set up conditions for energized work, and they assist the team in creating an informative workspace. • They help the team generate organizational trust and goodwill, and they often take responsibility for any reporting needed. • Coaches also help team members remember to maintain their self-discipline, helping them remain in control of challenging practices such as risk management, test-driven development, slack, and incremental design and architecture.	2	K2	CO2



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12.	Outline the types of Team Distribution in Agile • Co-located Teams • Distributed Teams • Hybrid Teams	2	K2	CO2
13.	Describe Epics in User Stories? Epics are large work items broken down into a set of stories, and multiple epics comprise an initiative. These larger structures ensure that the day-to-day work of the development team (on stores) contributes to the organizational goals built into epics and initiatives.	2	K2	CO2
14.	Summarize the benefits of User Stories. Stories keep the focus on the user. A to-do list keeps the team focused on tasks that need to be checked off, but a collection of stories keeps the team focused on solving problems for real users • Stories enable collaboration. With the end goal defined, the team can work together to decide how best to serve the user and meet that goal. • Stories drive creative solutions. Stories encourage the team to think critically and creatively about how to best solve for an end goal. • Stories create momentum. With each passing story, the development team enjoys a small challenge and a small win, driving momentum.	2	K2	CO2
15.	Define product backlog? A product backlog is a prioritized list of tasks — including new features to build, bugs to fix, improvements to implement, etc. that is derived from the product requirements and roadmap. It is a scrum artifact that serves as a to-do list for agile development teams.	2	K1	CO2
16.	Mention the accountable person for prioritizing the product backlog? Product manager The product manager is ultimately responsible for the product backlog's content and prioritization. However, that doesn't mean that PMs must work alone. Great professionals set the right context and let the team evolve the product backlog content. Stakeholders Many teams encourage stakeholders to create tickets in the product backlog. That often gets in the way of collaboration because it becomes a service-provider relationship. Agile teams : Team members are encouraged to create product backlog items. For example, software engineers commonly create	2	K1	CO2



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	tech debt items and designers point out usability glitches. That said, it's the product manager's responsibility to decide what makes it to the top of the list.			
17.	State the release plans in Agile? Agile release planning (sometimes called Scrum release planning) is where teams build and release products in iterations or incremental stages rather than all at once or grouped into distinct manor releases.	2	K1	CO2
18.	Classify the elements of a product release map. • The proposed release(s) for the project • Plans for each release • Subsequent iterations for the release(s) • Plans for each iteration • Feature development within an iteration • Individual tasks necessary to deliver a feature	2	K2	CO2
19.	Describe the phases of Risk Management process. ○ Risk Identification ○ Risk Analysis ○ Risk Response ○ Monitoring and Control	2	K2	CO2
20.	How does improper estimation affect sprint planning? Improper estimation leads to either overloading or underutilizing the team. Over estimation causes missed commitments, while underestimation reduces productivity, both impacting sprint goals and team morale.	2	K2	CO2



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PART B

Q. No	Question	Marks	BTL	CO
1.	Explain the structure and roles of Scrum teams, XP teams, and general Agile teams.	16	K2	CO2
2.	A project is distributed across multiple locations. Apply Agile practices to manage team distribution and ensure collaboration.	16	K3	CO2
3.	Illustrate in detail about the Phases of Agile Project Lifecycle	16	K2	CO2
4.	An organization is starting a new product. Demonstrate how to create a product vision and translate it into a product backlog.	16	K3	CO2
5.	A project has many features but limited time. Apply prioritization and estimation techniques to create a release plan.	16	K3	CO2
6.	Explain how a product backlog is created and maintained in Agile projects.	16	K2	CO2

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Course Code	:	JAL2006
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No. of. Hours	:	9
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PART A

Q. No	Question	Marks	BTL	CO
1.	List the three main Scrum artifacts? The three main Scrum artifacts are: 1. Product Backlog 2. Sprint Backlog 3. Increment	2	K1	CO3
2.	Describe Product Backlog in Scrum The Product Backlog is an ordered list of everything that is known to be needed in the product, maintained by the Product Owner.	2	K2	CO3
3.	How does Sprint Backlog will work? The Sprint Backlog is a set of Product Backlog items selected for the Sprint, along with a plan for delivering the product Increment and achieving the Sprint Goal.	2	K2	CO3
4.	Define Increment in Scrum? An Increment is the sum of all Product Backlog items completed during a Sprint and the value of previous Increments, representing a step toward the product goal.	2	K1	CO3
5.	Mention the responsible person for managing the Product Backlog? The Product Owner is responsible for managing and prioritizing the Product Backlog.	2	K1	CO3
6.	Summarize the role of the Developers in Scrum? Delivering increments: They build and deliver a functional, potentially releasable product increment each Sprint. Planning the work: They participate in Sprint Planning and decide how to turn Product Backlog items into a usable increment. Maintaining quality: They ensure the product meets the agreed Definition of Done .	2	K2	CO3



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	Self-organizing: They manage their own work without being directed by others. Continuous improvement: They adapt and improve their processes during Sprint Retrospectives. Developers are responsible for delivering a potentially releasable Increment of the product at the end of each Sprint.			
7.	Discuss the Scrum Master's responsibility? The Scrum Master ensures that Scrum is understood and enacted and helps remove obstacles to the team's progress.	2	K2	CO3
8.	Mention the importance of Product Planning in Scrum? Product Planning involves defining the vision, roadmap, and high-level features to guide the development of the product over time.	2	K1	CO3
9.	Illustrate the purpose of a Scrum Team simulation exercise? It helps participants experience Scrum roles, events, and collaboration in a real-time, risk-free environment.	2	K2	CO3
10.	Why the product backlog remain dynamic throughout the project? The product backlog evolves to reflect changing requirements, customer feedback, and market conditions. This flexibility ensures that development remains aligned with business goals and delivers relevant, high-value features.	2	K2	CO3
11.	How does a Scrum simulation help in understanding team dynamics? It allows participants to practice communication, collaboration, and problem-solving as a team under Scrum principles.	2	K2	CO3
12.	Why is time-boxing important in a Scrum simulation? It mimics real Scrum Sprints, encouraging focus, discipline, and time management.	2	K2	CO3
13.	How does a Burndown Chart help in Sprint management? It helps by - Tracking daily progress of remaining work in the Sprint - Identifying delays early if work is not reducing as expected - Supporting decision-making, such as adjusting tasks or effort - Improving transparency for the team and stakeholders	2	K2	CO3



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14.	Differentiate between Sprint planning and release planning. Sprint planning focuses on short-term work for a Sprint, while release planning focuses on long-term delivery of features.	2	K2	CO3
15.	Define Product Roadmap? A Product Roadmap is a high-level, time-based visual summary that outlines the product's direction and major deliverables.	2	K1	CO3
16.	What happens if a team member has an issue or blocker in the Daily Scrum? The issue is flagged, and the Scrum Master or team members can help address it after the meeting, outside the Scrum.	2	K1	CO3
17.	How does velocity help in Release Planning? Velocity helps predict how much work the team can complete in future Sprints, aiding in more accurate release forecasts.	2	K2	CO3
18.	Outline the main purpose of a Sprint Review? To inspect the increment and adapt the product backlog if needed.	2	K2	CO3
19.	How does the Daily Scrum improve communication? The Daily Scrum in Scrum improves communication by providing a regular and structured opportunity for team members to share updates. • Team members discuss their progress and upcoming work • They highlight any issues or blockers • It ensures everyone is aligned with the Sprint Goal • It promotes quick information sharing and coordination	2	K2	CO3
20.	Mention the factors affect release planning? Factors include • Team velocity • Backlog size • Priorities • Deadlines.	2	K1	CO3



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PART B

Q. No	Question	Marks	BTL	CO
1.	Discuss the various roles and activities in Scrum. How do they contribute to the success of a Scrum project?	16	K2	CO3
2.	Describe Scrum artifacts such as Product Backlog, Sprint Backlog, and Increment in detail.	16	K2	CO3
3.	A team faces confusion in daily coordination. Apply Scrum meetings to improve communication and workflow.	16	K3	CO3
4.	Explain the Sprint cycle including planning, execution, review, and retrospective.	16	K2	CO3
5.	A team needs to ensure quality before delivery. Apply acceptance testing to verify user stories.	16	K3	CO3
6.	Apply the Burndown Chart technique to monitor progress and recover a delayed project.	16	K3	CO3

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PART A

Q. No.	Question	Marks	BTL	CO
1.	Name the role of the XP team in Extreme Programming (XP)? The XP team works collaboratively to deliver high-quality software through continuous feedback, iterative development, and close communication with the customer.	2	K1	CO4
2.	Illustrate the primary roles in an XP team? The primary roles are the Customer, Developer, and Coach.	2	K2	CO4
3.	Outline the key principle that guides an XP team's collaboration? Communication and feedback are the core principles, ensuring close collaboration and alignment with customer needs.	2	K2	CO4
4.	Mention the role of the Customer in an XP team? The Customer defines user stories, provides feedback, and clarifies requirements during the development process.	2	K1	CO4
5.	How does pair programming work in an XP team? Pair programming involves two developers working together at the same computer, one writing the code while the other reviews and provides guidance.	2	K2	CO4
6.	State the purpose of continuous integration in XP? To ensure that code is integrated frequently, typically multiple times a day, to catch issues early and maintain a working codebase.	2	K1	CO4
7.	Describe the "Collective Code Ownership" in XP? It means that every team member is responsible for the entire codebase, and anyone can improve or modify any part of the code at any time.	2	K2	CO4
8.	How does an XP team handle changes in requirements during a project? XP teams welcome changing requirements, even late in the	2	K2	CO4



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	development process, through regular feedback and iterative development.			
9.	Mention the purpose of the "Sustainable Pace" in XP? To ensure that the team works at a pace that can be sustained over time, preventing burnout and maintaining long-term productivity.	2	K1	CO4
10.	Illustrate why simplicity important for an XP team? Simplicity in design and implementation ensures that the codebase is easy to maintain, extend, and understand, reducing complexity over time.	2	K2	CO4
11.	Describe "code smell," and how is it related to refactoring? Code smell refers to any symptom in the code that indicates a potential problem, such as duplications or overly complex methods, which can be addressed through refactoring.	2	K2	CO4
12.	State whether refactoring introduce bugs into the software? Yes, if not done carefully or without proper testing, refactoring can introduce bugs, which is why thorough testing is necessary after refactoring.	2	K1	CO4
13.	Define "driven development," and how does it relate to refactoring? Test-Driven Development (TDD) encourages writing tests before code and refactoring safely by ensuring that all changes maintain existing functionality through tests.	2	K1	CO4
14.	Specify the role do automated tests play in refactoring? Automated tests help ensure that refactoring doesn't break existing functionality by validating that the software still works after changes are made.	2	K1	CO4
15.	Classify some common refactoring techniques? Common techniques include renaming variables or methods, extracting methods, eliminating duplication, and simplifying conditional expressions.	2	K2	CO4
16.	How does velocity help with sprint planning in XP? Velocity gives the team an estimate of how much work they can handle in the next iteration based on past performance.	2	K2	CO4
17.	What does it mean if a team's velocity is inconsistent in XP? Inconsistent velocity may indicate issues with team collaboration, understanding of requirements, or other factors impacting team performance.	2	K2	CO4



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18.	Summarize the benefits of pair programming? Benefits include improved code quality, knowledge sharing between team members, faster problem-solving, and better collaboration.	2	K2	CO4
19.	How does collective code ownership promote collaboration in XP? Collective code ownership ensures that all team members are responsible for the entire codebase, encouraging collaboration and reducing bottlenecks, as anyone can modify or improve any part of the code.	2	K2	CO4
20.	Specify the role of feedback in collaboration within XP? Feedback is essential in XP as it enables continuous improvement. Frequent feedback from peers, customers, and automated tests helps the team collaborate and refine both the product and processes.	2	K1	CO4

PART B

Q. No	Question	Marks	BTL	CO
1.	Describe the concept of technical debt in XP and their importance.	16	K2	CO4
2.	Illustrate in detail about Pre-requisites, Challenges in adopting XP .	16	K2	CO4
3.	Apply velocity concepts to plan iterations for a software project.	16	K3	CO4
4.	A team struggles to complete tasks within deadlines. Demonstrate how timeboxing can improve productivity and time management.	16	K3	CO4
5.	How does collaboration contribute to high-quality software development, and what practices are most effective in fostering collaboration among team members, customers, and stakeholders?	16	K2	CO4
6.	A project is delayed due to repeated debugging. Apply pair programming techniques to minimize errors and speed up development.	16	K3	CO4

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Regulation	:	2023
Course Name	:	AGILE SOFTWARE DEVELOPMENT
Course Code	:	JAL2006
Year/Semester/Branch	:	II/IV/AI&ML
Unit No	:	V
Unit Title	:	AGILE SOFTWARE DESIGN AND DEVELOPMENT
No. of. Hours	:	9
Weightage in Exam(%)	:	20

PART A

Q. No.	Question	Marks	BTL	CO
1.	Mention the main purpose of using design principles in software development? To create software that is maintainable, scalable, and easy to understand.	2	K1	CO5
2.	How do design principles help during software maintenance? They reduce complexity, making the code easier to modify and debug.	2	K2	CO5
3.	Why are design principles important in object-oriented design? They help in organizing code using clear, reusable, and modular structures.	2	K2	CO5
4.	Name any two key benefits of applying design principles. Improved code quality and easier testing.	2	K1	CO5
5.	How do design principles support Agile methodologies like XP? They allow for frequent refactoring and continuous delivery by keeping code clean and adaptable.	2	K2	CO5
6.	Indicate the impact of ignoring design principles in a project? It leads to code that is hard to maintain, extend, and prone to bugs.	2	K1	CO5
7.	How do design principles promote code reusability? By encouraging modular and loosely coupled components.	2	K2	CO5
8.	State the role of design principles in team collaboration? They provide a common structure and best practices, helping developers work consistently.	2	K1	CO5
9.	How do design principles support the concept of "Separation of Concerns"? They ensure each part of the software handles a specific responsibility, improving clarity.	2	K2	CO5



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10.	Why are design principles essential for long-term project success? They reduce technical debt and make future enhancements easier and safer.	2	K2	CO5
11.	Define Continuous Integration (CI)? Continuous Integration is a development practice where developers frequently integrate code into a shared repository, typically several times a day, and each integration is verified by an automated build and test.	2	K1	CO5
12.	Define code refactoring? Refactoring is the process of restructuring existing code without changing its external behavior to improve its readability, maintainability, and structure.	2	K1	CO5
13.	How does Agile design handle changing requirements? Agile design encourages adaptive planning and incremental development to easily accommodate changing requirements.	2	K2	CO5
14.	State the role of simplicity in Agile design? Simplicity ensures that only necessary features are built, reducing complexity and making the code easier to maintain.	2	K1	CO5
15.	Classify any two Agile design practices. Test-Driven Development (TDD) and Refactoring.	2	K2	CO5
16.	Summarize the three main steps of TDD? Red (write a failing test), Green (write code to pass the test), and Refactor (improve the code).	2	K2	CO5
17.	How does TDD improve code quality? TDD ensures that code is thoroughly tested, reduces bugs, and encourages modular, maintainable code design.	2	K2	CO5



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PART B

Q. No	Question	Marks	BTL	CO
1.	Discuss in detail about Design practices and role of design principles.	16	K2	CO5
2.	Explain in detail about refactoring techniques with examples.	16	K2	CO5
3.	A software development team is facing delays, manual errors, and difficulty in maintaining code quality due to manual processes. Apply Agile automated tools to improve development efficiency, testing, and delivery. Explain the steps, tools, and benefits in this scenario.	16	K3	CO5
4.	A software project team is facing frequent integration issues, late bug detection, and delayed releases. Apply Continuous Integration practices to solve these problems.	16	K3	CO5
5.	Explain in detail about Agile approach to Quality Assurance	16	K2	CO5
6.	A software team is facing frequent defects and unclear requirements. Apply Test Driven Development (TDD) practices to improve code quality and reduce errors. Explain the steps, implementation, and expected benefits in this scenario.	16	K3	CO5

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